

Course: Data Mining (COMP7/8118)

Student: Neehanth Reddy, Maramreddy

UID: U00924209

Date: 9/16/25

Project Proposal: E-Commerce Analytics: Customer Segmentation, Market Basket Mining, and Sales Forecasting

Problem/application:

An e-commerce business needs to move beyond simple sales reports to make strategic decisions. This project addresses three critical operational areas:

1. **Marketing Strategy:** A "one-size-fits-all" approach is inefficient. By segmenting customers, the business can design targeted campaigns that improve retention and maximize value.
2. **Sales & Merchandising:** Understanding which products are bought together can unlock opportunities for product bundling, cross-selling, and optimizing online store layout to increase the average order value.
3. **Operations & Finance:** Accurate sales forecasts are essential for managing inventory, planning logistics, and making sound financial decisions.

This project will deliver a multi-faceted analysis providing actionable insights for all three areas.

Methods will be tested or implemented:

The project will be implemented in **Python** using libraries like **Pandas**, **Scikit-learn**, and **Statsmodels**. It will consist of three distinct data mining components:

1. Customer Segmentation (Clustering):

- **RFM Feature Engineering:** Transform the transaction log into a customer-centric view by calculating **Recency**, **Frequency**, and **Monetary** values for each customer.
- **K-Means Clustering:** Use the scaled RFM features to group customers into distinct segments and analyze their characteristics to create personas (e.g., "Loyal Champions," "At-Risk Customers").

2. Market Basket Analysis (Association Rule Mining):

- **Data Transformation:** Convert the sales data into a transactional format, where each invoice is a "basket" of items.
- **Apriori or FP-Growth Algorithm:** Apply this algorithm to discover frequent itemsets and generate association rules (e.g., {Product A} -> {Product B}).

- **Rule Evaluation:** Analyze the generated rules using metrics like Support, Confidence, and Lift to identify strong and actionable product relationships.

3. Sales Forecasting (Time Series Analysis):

- **Data Aggregation:** Resample the sales data into a time series (e.g., daily or weekly total revenue).
- **SARIMA Model:** Implement a Seasonal AutoRegressive Integrated Moving Average (SARIMA) model to capture both the underlying trend and the seasonality in sales patterns.
- **Forecasting:** Train the model on historical data and use it to predict future sales over a defined period.

Dataset:

The project will use the "[Online Retail Data Set](#)" from the UCI Machine Learning Repository. This dataset is ideal as it contains over 500,000 transactional records from a UK-based online retailer, including customer IDs, invoice dates, quantity, and price – all the necessary components for an RFM analysis.

Potential challenges to implementation:

- **Parameter Tuning:** All three models rely on careful parameter tuning. This includes selecting the optimal number of clusters (k) for K-Means, setting support/confidence thresholds for association rule mining, and identifying the correct parameters (p,d,q, P,D,Q) for the SARIMA model.
- **Rule Interpretation:** The market basket analysis can generate thousands of rules. A key challenge will be filtering these to find patterns that are not only statistically significant but also novel and commercially interesting.

Expected Deliverables:

- GitHub repository URL
- Python source code (likely in a Jupyter Notebook)
- Project Report with comprehensive analysis